

Production Full-CMG Benchmarks for VCkss

Private 0.4.0-alpha.1 candidate versus maintained MATLAB KSS

VCkss development report

27 August 2026

Abstract

This report qualifies the ordinary-build `CMG_FULL_V2` route in VCkss against the maintained MATLAB KSS implementation on two registered hard problems. Both use 200 randomized projections and four application workers, with one cold and five position-balanced warm runs. At source `4b6874e`, VCkss’s median complete-command time is 0.716 times MATLAB on the macOS 8,192-firm case and 0.578 times MATLAB on fixed CZ18. Peak process-tree memory is 0.421 and 0.244 times MATLAB, respectively. Both cases pass the selected alpha gate—faster than matched MATLAB, no more than five percent slower than the source-bound private winner, and all statistical, complete-residual, memory, state, and lifecycle checks passing. The aspirational 2x MATLAB target is not claimed.

1 Decision and scope

The promoted implementation directly solves VCkss’s prepared hybrid worker–firm Laplacian. It vendors exact standalone CMG source `dbefbc5` and reuses one graph, hierarchy, Rayon pool, parallel plan, and admitted workspace pool across 601 fit, leverage, and target right- hand sides. It does not route exact results through JLA, does not use the retired simplified hierarchy, and does not select a fallback after estimator RNG begins.

The production cell is deliberately narrow: no controls, weights, target weights, or explicit deletion ID; match deletion, joint nuisance, mover headline, JLA, automatic engine/preconditioner/batch, Counter-V1, and an effective observation key. It is available through explicit `backend(rust)` and, on qualified macOS/Linux runtimes, eligible `backend(auto)` requests. Explicit `backend(mata)` remains permanent. Windows, a public tag, and public distribution are outside this report.

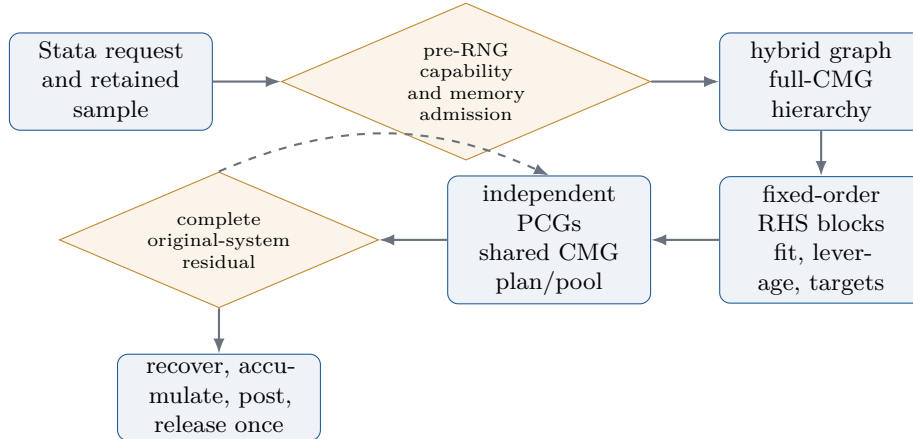


Figure 1: Production dataflow. Admission and the deterministic refinement schedule are frozen before Counter-V1. A failed selected route returns a typed error; it never changes estimator or backend after RNG.

2 Registered protocol

Item	Registered value
Production source	4b6874ededa1244bca389e4ee82148b42b9693a1
CMG source	dbefbc5e3b442c6dde6e7861a66d82fd5ed24f10
Toolchain	Rust 1.85.1; VCkss MSRV 1.85; VCkss edition 2021; vendored CMG edition 2024
Estimator	JLA, 200 probes, four workers, one cold plus five alternating warm runs
Tolerances	fit 10^{-10} ; randomized probes 10^{-6} ; fit inner start 10^{-12}
Residual gate	complete original system $\leq \max(10^{-11}, 10\tau)$; same-route 10x refinement only
macOS	1,966,080 rows, 327,680 workers, 8,192 firms; Stata/MP 19; MATLAB R2024b Update 5
SCC	8,201,888 rows, 117,529 workers, 10,603 firms; Stata/MP 19; MATLAB R2024b Update 3
Timing	complete estimator command; medians of five warm repetitions; launch order position-balanced
Memory	complete process-tree peak RSS; VCkss also receipts pre-RNG forecast, admitted peak, and retained bytes

The two inputs are fixed by SHA-256: macOS 19744b8418ffff82461527d7426cdc976dce18bf36597555d19bf847dec5afbe and SCC 1748ca2a6a46f248e05c0329407e7e7708ec7628c1ffce5f0e06ee264bdf0575. MATLAB source and runtime trees are separately hashed. Queue wait, source deployment, compilation, MATLAB pool startup, and MEX setup are reported but excluded from the complete estimator command in both applications.

3 End-to-end performance

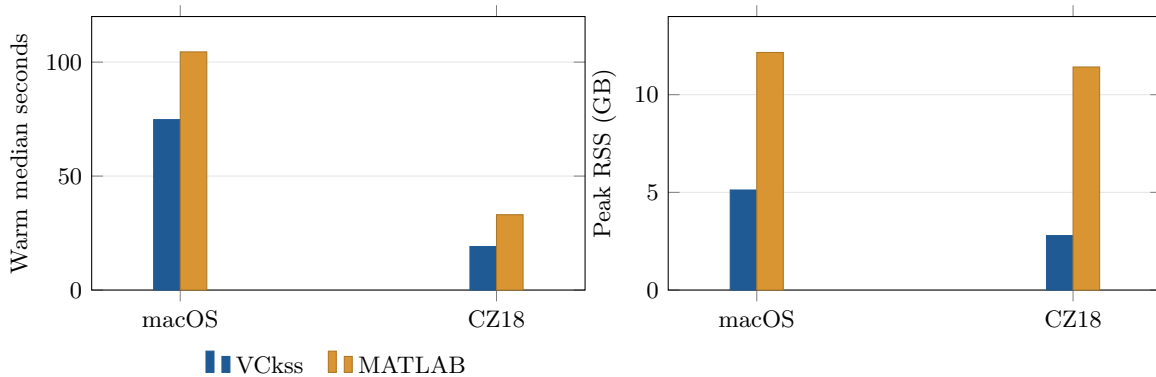


Figure 2: Median complete-command time and process-tree peak RSS at four application workers. Lower is better.

Table 1: Warm-median promotion results.

Case	VCkss (s)	MATLAB (s)	Ratio	Speed advantage	Private (s)	Prod./private
macOS 8,192-firm	74.774	104.489	0.716	1.397x	81.145	0.921
SCC fixed CZ18	19.097	33.058	0.578	1.731x	33.942	0.563

Both selected promotion rules pass. Neither case regresses from its private winner; production is faster in both. The macOS ratio is above the 0.5 stretch target, and the SCC result remains above the 0.5 threshold. No 2x release claim is made.

4 Where command time goes

Repeated independent solves are still the largest macOS phase. On CZ18, Stata/native preparation and data movement are comparably important after the faster, smaller Laplacian solve. The shortest next optimization path

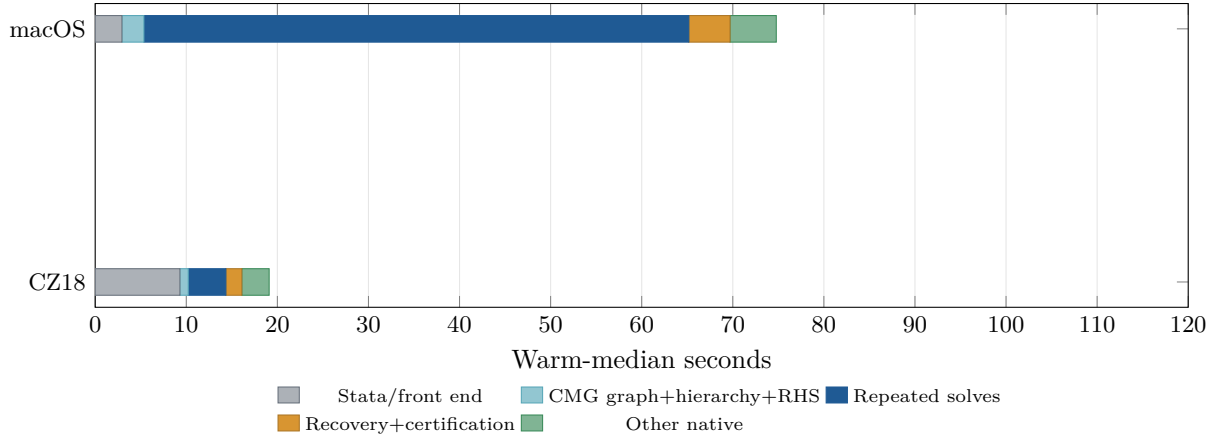


Figure 3: A reconciled decomposition of complete-command medians. Categories sum to the reported command median; “other native” is the receipted native solve remainder after the separately timed full-CMG phases.

Table 2: Detailed median native and full-CMG receipts (seconds).

Phase	macOS	CZ18	Phase	macOS	CZ18
Native ingest	0.032	0.245	Full-CMG graph	0.428	0.055
Canonicalization	0.573	1.719	Hierarchy/plan	0.133	0.015
Native graph	0.447	0.546	RHS construction	1.850	0.844
Compression	0.522	1.397	Repeated solve	59.826	4.131
Native plan	0.152	1.019	Extract/recover/certify	4.523	1.755
Native solve total	71.841	9.799	Complete command	74.774	19.097

is therefore platform/case specific: improve repeated full-CMG applications on the synthetic headline while profiling front-end preparation on CZ18. Retired fused, mixed-precision, and pass-fused prototypes remain disabled because they did not beat the scalar winner under the registered end-to-end rule.

5 Solver work, residuals, and statistical outputs

Table 3: Warm-median full-CMG work and numerical certification.

Measure	macOS	SCC CZ18
Right-hand sides	601	601
PCG iterations	4,817	7,001
Operator applications	5,418	8,886
CMG applications	4,817	7,001
Refinement attempts / columns	0 / 0	35 / 1,284
Maximum reduced residual	7.862×10^{-6}	1.541×10^{-3}
Maximum complete residual	6.971×10^{-6}	9.975×10^{-6}
Default probe limit	10^{-5}	10^{-5}

CZ18’s attempt-level reduced maximum may exceed the final threshold because the receipt retains the first attempt. The frozen schedule deterministically re-solves only failing columns at a 10x tighter inner tolerance. Every selected column passes the independent complete original worker-plus-firm system before posting. Refinement never changes RNG draws, hierarchy, result family, or backend.

MATLAB uses a different RNG, solver contract, and, on CZ18, target-weight convention. It does not emit a compatible numerical MCSE. Its point differences therefore document statistical agreement but are not falsely presented as pathwise equality. The common-probe private comparisons pass the registered equivalence rule.

Table 4: Corrected-target comparison. MATLAB differences are descriptive; private-winner differences use common Counter-V1 probes and are release-blocking.

Target	Absolute VCkss–MATLAB difference		Absolute production–private difference	
	macOS	CZ18	macOS	CZ18
Worker variance	6.19×10^{-6}	1.48×10^{-3}	0	2.20×10^{-9}
Firm variance	1.44×10^{-5}	7.63×10^{-5}	0	3.67×10^{-9}
Covariance	1.11×10^{-5}	1.07×10^{-4}	0	2.45×10^{-9}
Total	1.64×10^{-6}	1.62×10^{-3}	0	9.75×10^{-10}

Data, caller RNG, sort RNG, `e(sample)`, accounting, target identity, typed status, and idle-registry checks pass in every accepted run.

6 Memory, cancellation, and lifecycle

Table 5: Warm-median and receipted memory (GB, decimal).

Measure	macOS	SCC CZ18
Pre-RNG conservative forecast	32.828	9.788
Admitted full-CMG peak	19.500	7.133
Actual retained native storage	1.184	0.095
Observed VCkss process-tree RSS	5.121	2.788
Observed MATLAB process-tree RSS	12.161	11.417

The forecast covers retained input, canonicalization, raw match preparation, hybrid and CMG graph copies, hierarchy bounds, parallel plan, thread/workspace pool, RHS/results, extraction, residual checks, ABI copies, transients, and an allocator allowance. Graph and hierarchy construction completes before Counter-V1, actual retained sizes are reconciled to the forecast, and a failed admission returns without estimator RNG.

Owned work runs under a coordinator worker. The Stata caller thread polls UserBreak and sets an atomic cancellation token observed by preparation, CMG, PCG, recovery, and certification code; Rayon workers never call the Stata API. Success, memory rejection, cancellation, convergence failure, corrupt receipt, posting failure, panic, and stale-handle tests require one terminal generation, exactly-once release, and an idle registry.

7 Platform qualification and provenance

The same runtime source, packaged at macOS receipt tip `4dafec6734af4b8d3c25785f268f19f69f780684`, passed macOS arm64 and x86-64 thin plugins, the universal plugin, Rosetta execution, clean installation, Rust/C/ABI checks, full-CMG explicit and automatic routing, cancellation, memory/refinement, and lifecycle coverage. SCC job 7330577 used Linux qualifier source `992eba0947ca155c534f532750fc202e41ecf978`, built a normal Linux x86-64 plugin from the deterministic bundle, and ran the locked Rust, C, ABI, Stata full-suite, and clean-install gates under Stata/MP 19. Its wrapper and `qacct` records report `failed=0` and `exit_status=0`.

The local source-bound supply-chain gate at receipt commit `4dafec6` audits the workspace, Stata-backend, and fuzz lockfiles with `cargo-audit 0.22.2`; zero vulnerabilities and warnings are accepted. Four deterministic CycloneDX 1.5 SBOMs record 53 components, licenses, package URLs, and checksums. Human license/provenance approval remains required before any public distribution.

8 Acceptance and limitations

Gate	Status	Evidence
Faster than matched MATLAB on both registered cases	PASS	ratios 0.716, 0.578
No more than 5% slower than private winner	PASS	ratios 0.921, 0.563
Complete original-system residuals	PASS	within 10^{-5} in both cases
Corrected-target/common-probe policy	PASS	all four targets, both cases
Declared memory and RSS rule	PASS	no budget breach; RSS below MATLAB
macOS arm64/universal/Rosetta and SCC Linux	PASS	exact-source receipts
Windows and public release	DEFERRED	no claim, binary, tag, or public release
2x MATLAB development target	DEFERRED	directional target, not alpha gate

These are two registered hard cases, not a universal performance guarantee. The machines differ, MATLAB retains native implementation choices, and five warm repetitions support engineering medians rather than formal confidence intervals. The report does not claim econometric standard errors: Vckss posts point estimates and numerical Monte Carlo error only, with no $e(V)$.

9 Reproduction commands

All commands were run from a clean `main` checkout. The exact paths in the receipts are retained; portable forms are:

```
rust/stata_backend/qualify_macos.sh \
  --receipt /private/tmp/vckss-alpha-4dafec6-macos-receipt.txt \
  --artifacts-dir /private/tmp/vckss-alpha-4dafec6-macos-artifacts

/opt/anaconda3/bin/python3 vckss/benchmarks/full_cmg_production/run_local.py \
  --output-dir /private/tmp/vckss-full-cmg-alpha-4b6874e \
  --input-csv /private/tmp/vckss-local-matrix-787327f/run/input/input.csv \
  --matlab-root /Users/johannes/Git/varcomp_hdfe/Monte_Carlo/LeaveOutTwoWay

rust/stata_backend/scc/deploy_linux_bundle.sh REPO RUN_ID
ssh scc RUN_DIR/source/rust/stata_backend/scc/submit_linux_qualifier.sh RUN_DIR

vckss/benchmarks/full_cmg_production/submit_scc_cz18.sh \
  20260827T023000Z-alpha-cz18-4b6874e
```

Compact exact-source receipts, validator output, source and input hashes, platform identities, wrapper records, and scheduler accounting live under `vckss/benchmarks/full_cmg_production/evidence/` and `rust/qualification/evidence/`. Raw SCC runs remain under `/projectnb/welfgr/vckss/runs/`; disposable build/log caches are excluded from the maintained source tree. The final protected cleanup removed 26,310 regenerable files (4.187 GB) without touching any tracked receipt or report.